GIS

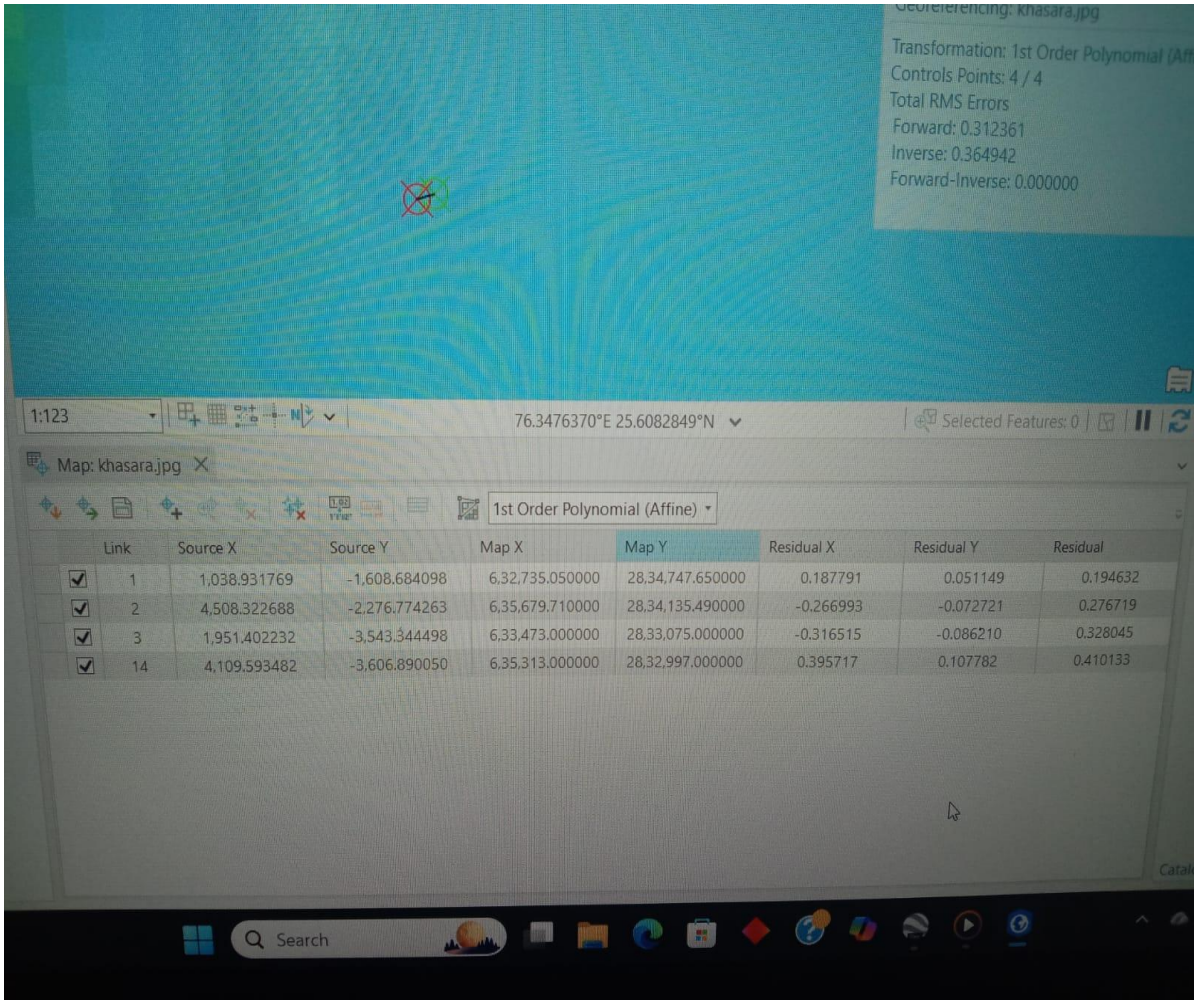
Khasra parcel project Workflow

Using ArcGIS PRO

Prepared by

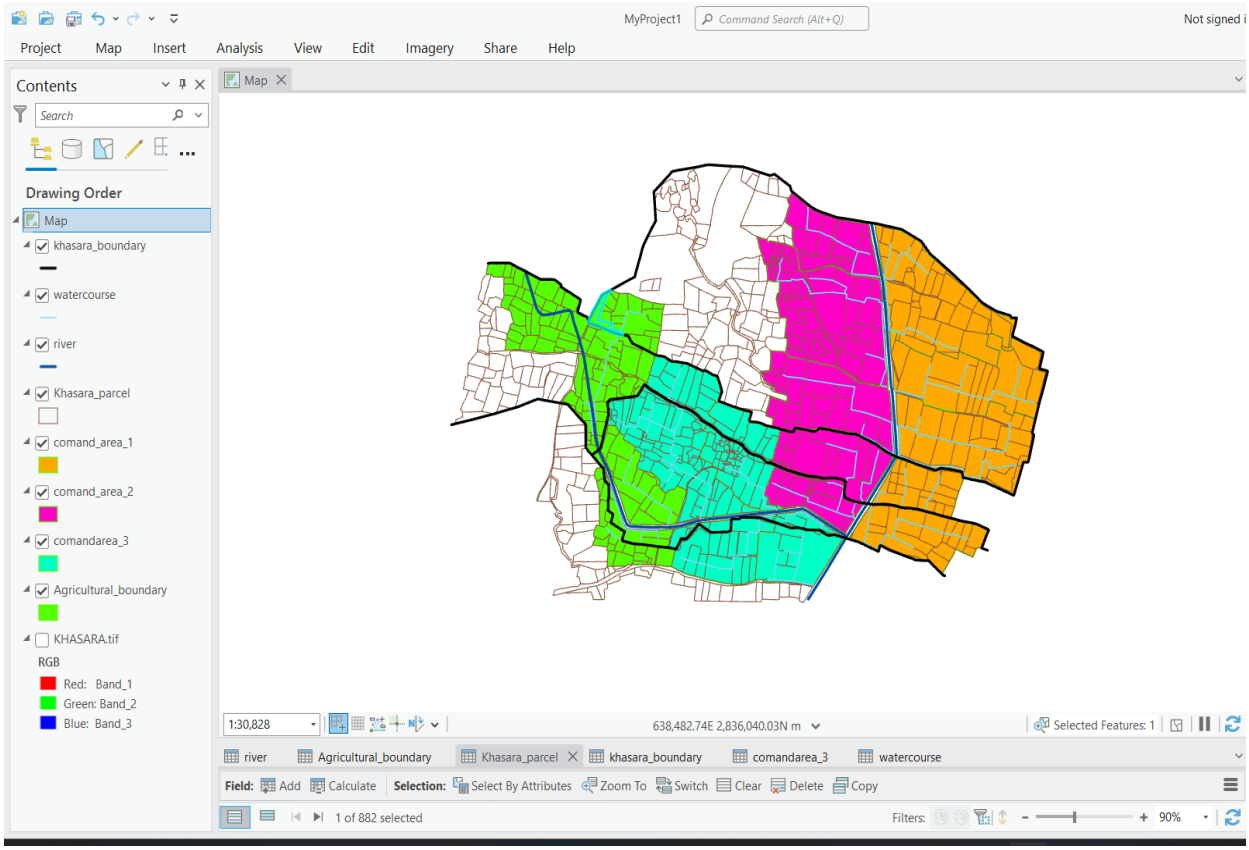
Sindhu. S

Georeferencing picture



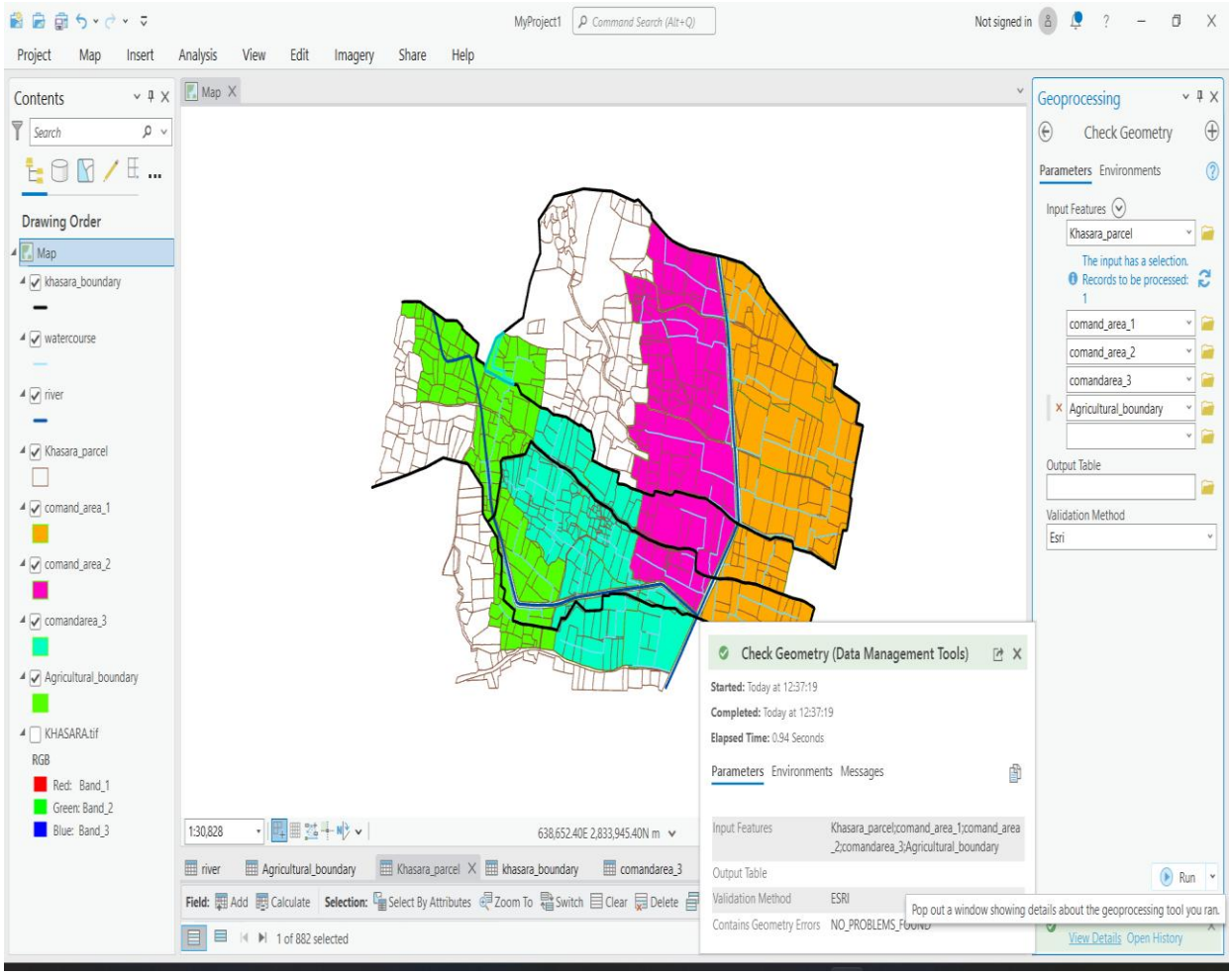
"Georeferencing was performed using 4 Ground Control Points (GCPs) based on the reference base map in the WGS 1984 UTM Zone 43N coordinate system. The RMS error values were maintained below 0.5, indicating high spatial accuracy for digitization and further analysis."

Digitization



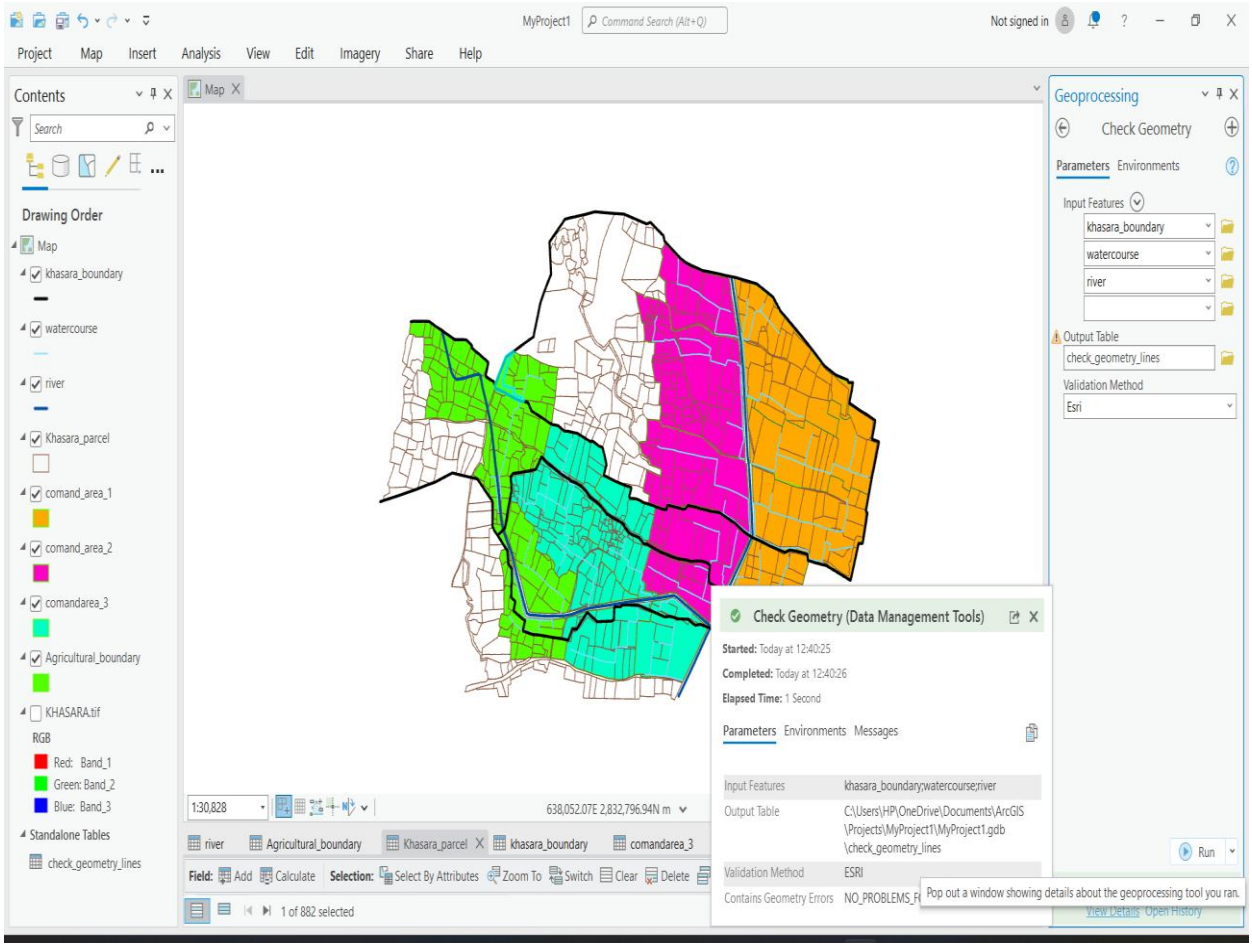
“Digitization was carried out based on the georeferenced base map. The workflow included digitizing major features such as Khasra boundary, river, watercourse, and agricultural boundary, followed by command area divisions (1, 2, and 3). Finally, detailed Khasra parcel boundaries were created to ensure accurate cadastral mapping. Snapping and editing tools were used to maintain spatial accuracy during digitization.”

Geometry Check for Polygon Feature



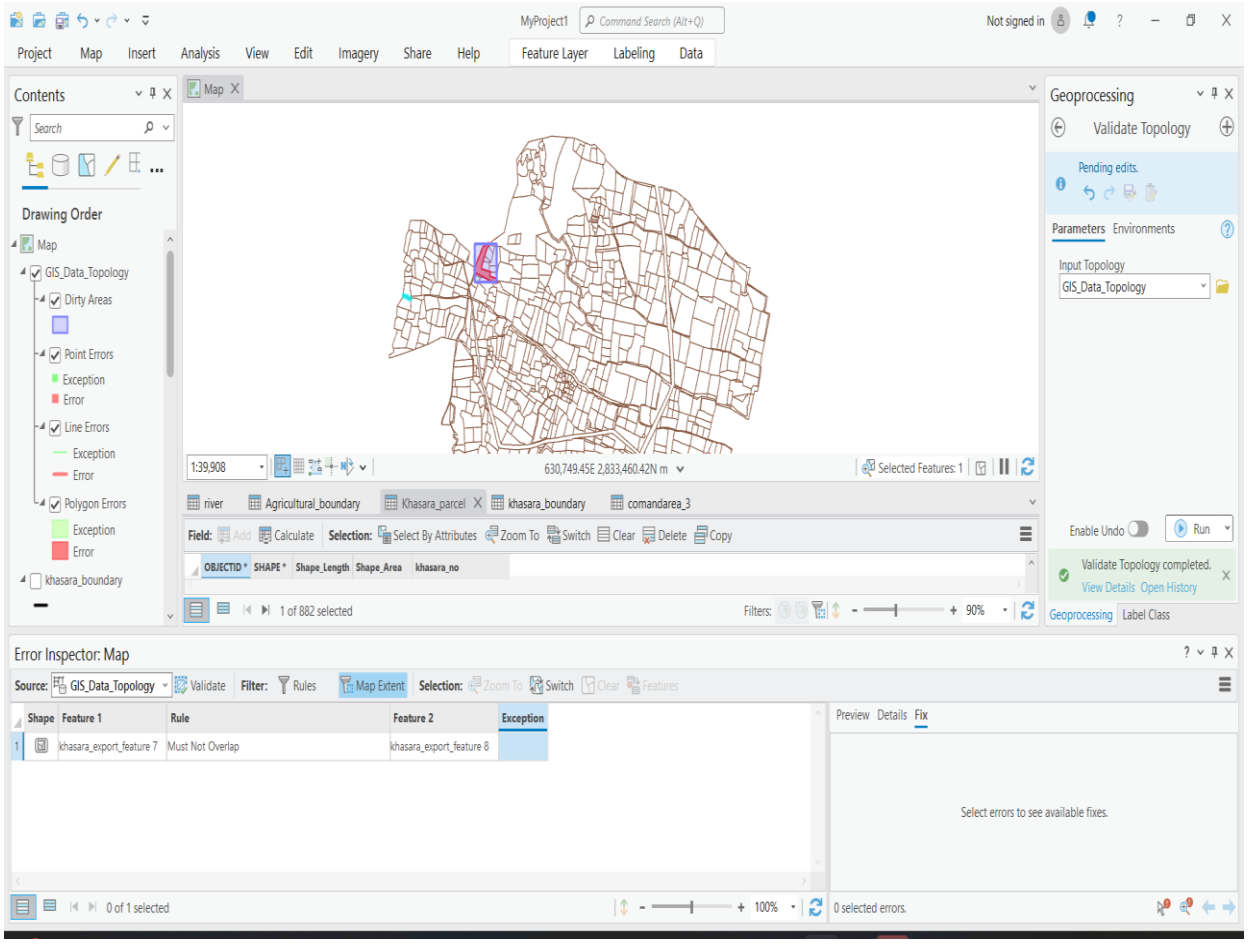
“Geometry check was performed on polygon features, including Khasra parcels and command areas (1,2,3), Agricultural boundaries to identify errors such as gaps, overlaps, and invalid geometries. No geometry errors were detected, indicating that the dataset is clean and spatially accurate for further topology validation.”

Geometry Check for Line Features



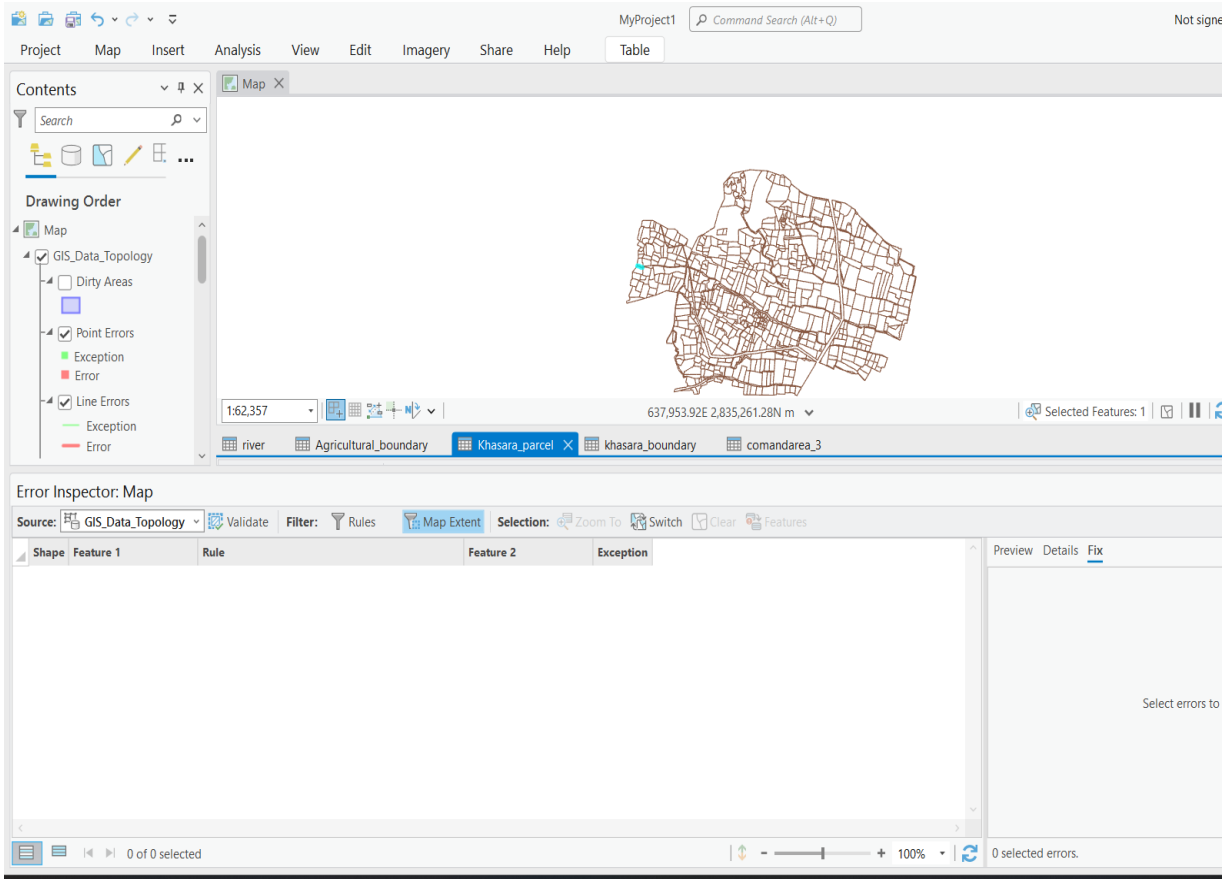
"Geometry check was performed on line features including Khasra boundary, river, and watercourse to identify errors such as dangles, overlaps, and invalid geometries. No geometry errors were detected, indicating that the line features are clean and topologically correct for further processing."

Topology Validation Showing Errors



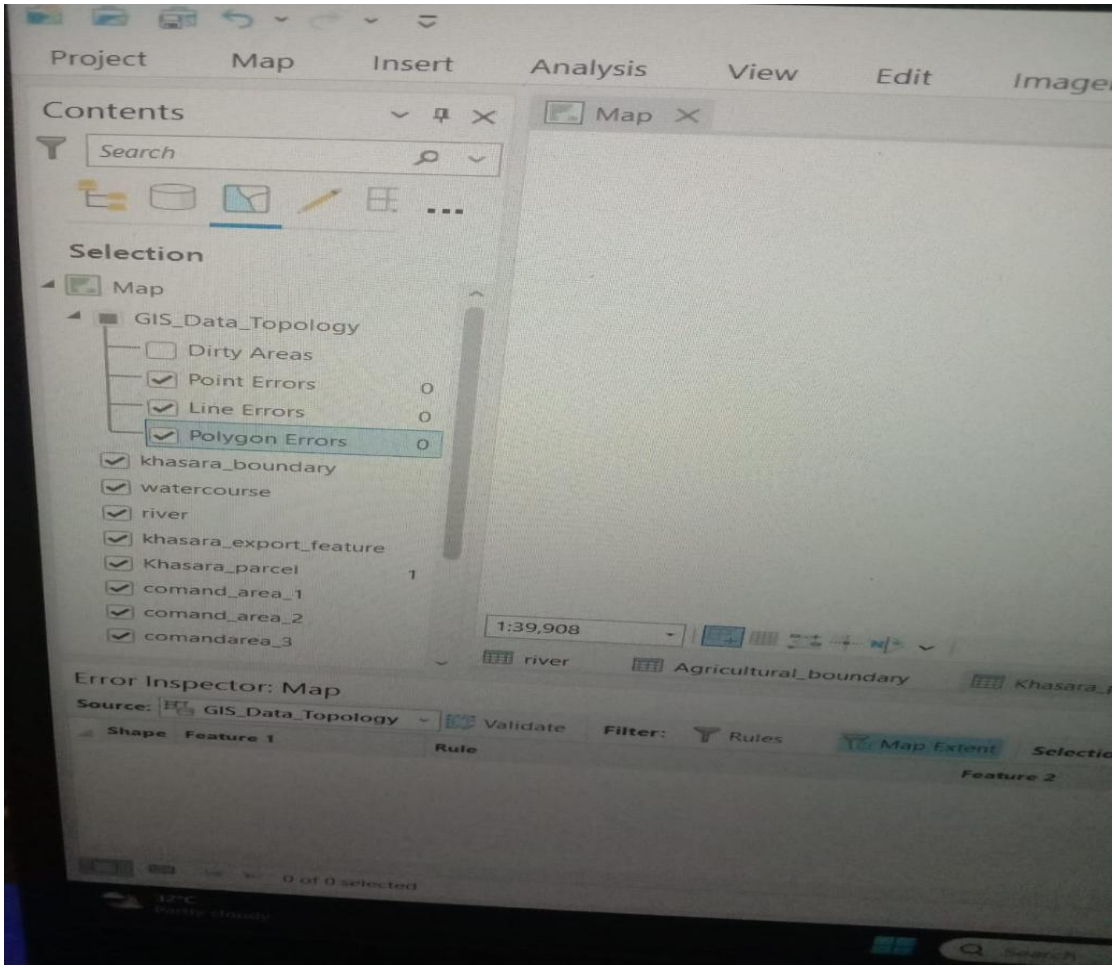
"Topology validation was performed by applying rules such as 'Must Not Overlap' and 'Must Not Have Gaps' on the Khasra parcel layer. Overlap errors were detected, while no gap errors were found, indicating that the parcel coverage is complete without missing areas."

Correction of Overlap Error



"The overlap errors identified during topology validation were corrected using editing tools. Appropriate methods such as 'Remove Overlap' and feature editing were applied to adjust parcel boundaries and eliminate overlapping areas, ensuring accurate and consistent spatial data. After correction, the dataset was revalidated to confirm that all errors were resolved."

Final Topology Validation



"Final topology validation results show that there are no dirty areas, point errors, line errors, or polygon errors in the dataset. This confirms that the spatial data is clean, accurate, and fully validated according to the defined topology rules."

Conclusion

“The Khasra parcel mapping project was successfully completed using ArcGIS Pro. The workflow included georeferencing, digitization, geometry validation, topology creation, and error correction. Final validation results confirmed zero errors, with no gaps, overlaps, or dirty areas. The dataset is accurate, topologically correct, and ready for mapping and analysis.”

“This project demonstrates strong skills in GIS data creation, validation, and quality control.”

Thank You

For Your Time and Consideration